CS315 · ALGORITHMS ANALYSIS & DESIGN

What Repeats

An analysis of five midterms: 472, 462, 461, 451 and 442

Muharib

What's inside

01 The bottom line

02 The repetition table

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01 The bottom line

Five midterms. Four of them (472, 462, 461 and 451) share nearly the same structure: a hash table question, a true/false and MCQ question, two dynamic programming or greedy questions, and a B-tree question.

Repeated word for word	Where
LCS: X = ACBAED, Y = ABCADF	472 · 462 · 451
Matrix chain: A=4×5, B=5×8, C=8×3, D=3×7	472 · 451
Hash table: [4, 13, 49, 6, 40, 33], size 9	472 · 451
T/F: Huffman, greedy solves all DP, and the MERGESORT equation	472 · 462 · 451
MCQ: the algorithm that uses past results	472 · 462 · 451
B-tree of order 5 with nearly the same numbers	472 · 451

So

The LCS question came with the same two strings in three midterms, and its answer is `ACAD` or `ABAD` with length 4. Learn the method because the instructor may change the strings, but start your revision from these questions.

02 The repetition table

The number in a cell is the question's mark. A ✓ means it appeared but the mark is

unknown, or it appeared as an item inside a question.

Topic	472	462	461	451	442	Lecture
Hash table: chaining, linear probing or double hashing	✓	5	5	4	✓	3
True / False + MCQ	✓	5	9	5	4	1 – 6
Longest common subsequence	✓	5	✓	4	—	4+
Matrix chain multiplication	✓	—	6	4	—	4+
Greedy knapsack (fractional)	—	5	5	—	—	4+
B-tree: build, insert and delete	✓	5	—	6	—	4+
Counting operations of **arrayMax**	—	—	—	2	2	1
Huffman: T/F, a code word or the running time	✓	✓	✓	✓	—	4+
Heaps, BST, trees and graphs	—	—	—	—	✓	—

"4+" means a lecture after the third.

03 Where are the marks?

Midterm	From Lectures 1 to 3	From the later lectures
462	About 7.5 of 25	About 17.5
461	About 9 of 25	About 16
451	About 7 of 25	About 18

Watch out

In every midterm, Lectures 1 to 3 carry only about a third of the marks. The other two thirds come from dynamic programming (LCS and matrix chain), greedy (knapsack, Huffman and activity selection) and B-trees.

04 Answers to repeated questions

Question	Answer
$T(n) = T(\frac{n}{2}) + T(\frac{n}{2}) + T(\frac{\text{MERGE}}{2})$	**False**
The recursive activity-selection problem algorithm takes a time of O(n log(n))	**Other value**: O(n)
An algorithm which uses the past results and uses them to find new results	**Dynamic programming**
f(n) is ω(g(n)) if is asymptotically ... than g(n)	**Strictly greater**
Best-case time complexity of linear search	**O(1)**
B-tree of degree 3	At most 3 children and 2 keys

05 Study plan

1 From Lectures 1 to 3

- Hash tables with all three methods. They appeared in all five. Practise with the Practice 2 file.
- The **arrayMax** counting table: `7n - 2`, and `7n` with the Temp line.
- True/false statements from the slides' wording: the definition of an algorithm, insertion sort "about n^2", the dictionary operations, and the log file.

2 The heavy questions

- LCS: diagonal + 1 when the letters match, otherwise the larger of top and left.
- Matrix chain: the cost is `p·q·r`, and fill the table from short chains to long.
- Fractional knapsack: sort by value ÷ weight and take in that order.
- B-tree: split when a node exceeds its maximum number of keys, and the middle key moves up.

All of them are solved step by step in the 462, 461 and 451 solution files.

3 The last day

Solve the 462 and 451 midterms in full, each against a 90-minute timer.